

RewLo

Technical Documentation

Stack · Project Structure · Configuration Guide

Prepared for new collaborators

Generated: Tue Jul 21 2026

```
artifacts-monorepo/
```

1. Project Overview

```
% % % artifacts/ % % % mobile/ !• Expo React Native app (@workspace/mobile)
```

RewLo is a sports fan digital wallet mobile app built with React Native and Expo. It lets fans connect to their favourite sports club, earn reward points when they pay at partner merchants, and redeem those points at checkout via the "RewLo Pay" agentic commerce flow.

```
% % % api-client-react/ !• Generated React Query hooks
```

The app is currently in Early Access Preview — all payment transactions are simulated and no real money moves. The goal is to validate UX and submit a polished app to the Apple App Store.

```
% % % api-spec/ !• OpenAPI Spec + Orval codegen
```

```
% % % api-zod/ !• Generated Zod validation schemas
```

```
% % % db/ !• Drizzle ORM schema + migrations
```

```
% % % scripts/ !• Shared utility scripts
```

```
% % % pnpm-workspace.yaml !• Workspace config + dependency catalog
```

```
% % % tsconfig.base.json !• Shared TypeScript strict defaults
```

The project is a pnpm workspace monorepo. All packages share dependencies through a central catalog defined in pnpm-workspace.yaml.

```
% % % package.json !• Root task orchestration
```

Package naming

- All workspace packages use the @workspace/ prefix
- artifacts/* are leaf packages (not referenced in root tsconfig.json)
- lib/* packages are composite and emit type declarations for sharing

3. Mobile App (artifacts/mobile)

```
app/
```

```
% % % _layout.tsx !• Root layout, fonts, WalletProvider
```

```
% % % index.tsx !• Redirect !' onboarding or home
```

```
% % % login.tsx !• Email/password login (any non-empty creds work in preview)
```

Framework: React Native 0.81.5 + Expo SDK 54

```
% % % onboard-unity.tsx !• Club selection on first launch
```

Navigation: Expo Router v6 (file-based, like Next.js but for mobile)

Language: TypeScript 5.9 — strict mode enabled

```
% % % score.tsx !• Live day live score / offers screen
```

State: React Context (WalletContext.tsx) OR no external state library

```
% % % checkout.tsx !• "RewLo Pay" agentic checkout flow
```

Fonts: Inter via @expo-google-fonts/inter

Icons: Expo Vector Icons (Ionicons) + expo-symbols (SF Symbols on iOS)

```
% % % _layout.tsx !• Tab bar config
```

Animation: React Native Reanimated 4 + Gesture Handler 2

```
% % % home.tsx !• Wallet balance, recent transactions, offers
```

Camera: expo-camera (QR code scanning for merchant checkout)

```
% % % activity.tsx !• Full transaction history
```

Architecture: New Architecture enabled (Fabric renderer + JSI)

```
% % % cards.tsx !• Linked payment cards
```

React Compiler: Enabled via babel-plugin-react-compiler (auto-memoisation)

```
% % % rewards.tsx !• Points & rewards catalogue
```

```
% % % profile.tsx !• Account settings, club, logout
```

Screen Structure (app/)

Key Source Files

- context/WalletContext.tsx — all app data (balance, points, transactions, cards, clubs)
- utils/agentCommerce.ts — RewLo Pay agent logic: step simulation, point recommendation
- constants/Colors.ts — brand palette (#00E5CC cyan, #041120 navy)
- app.json — Expo config (bundle ID, permissions, plugins)

Brand

Primary: #00E5CC (cyan)

Background: #041120 (navy)

Font: Inter (all weights)

UI Style: Dark mode only (userInterfaceStyle: "dark")

4. API Server (artifacts/api-server)

The API server is currently minimal — it serves as the backend foundation ready for future feature

```

    "scheme": "rewlo",
    "userInterfaceStyle": "dark",
    "newArchEnabled": true,
    "icon": "mobile/assets/images/rewlo-appicon.png",
    "splash": { "backgroundColor": "#041120" },
    "ios": {
      "bundleIdentifier": "com.rewlo.app",
    },
    "buildNumber": "1",
    "infoList": {
      "ITSAppliesNonExemptEncryption": false
    },
    "pin": "http://structured-JSON-logs-ever.console.log"
  },
  "build": esbuild('dist/index.mjs (CJS bundle)')
},
  "email": ReplSend (Replit integration installed)
}

```

Shared Libraries (lib/)

- lib/api-spec — OpenAPI 3 YAML spec; run codegen to regenerate clients
- lib/api-zod — Zod schemas and generators for the OpenAPI spec via Orval
- lib/api-client-react — React Query hooks auto-generated from the OpenAPI spec
- packages — Experiments:
 - "artifacts": {
 - "typedRoutes": true,
 - "lib/*": {
 - "reactCompiler": true

5. Key Configuration Files

app.json (Expo config)
pnpm-workspace.yaml

```
drizzle-orm: ^0.x
```

Defines which directories are workspace packages and pins shared dependency versions in a catalog so all packages use the same version without repeating version strings.

tsconfig.base.json (shared TypeScript)

- `strict: true` — full strict mode for all packages
- `incremental: true` — faster repeat typechecks via `.tsbuildinfo` cache
- `target: ES2022`, `module: NodeNext`
- `lib/*` packages set `composite: true` and emit declarations
- `artifacts/*` use `--noEmit` only (no declaration files emitted)

6. Routing & Service Proxy

- ## 6. Routing & Service Proxy
- Always access via `localhost:80` in `curl` / ad-hoc requests
 - Never call service ports directly (e.g. `localhost:5000`)
 - Paths are NOT rewritten – services handle their full base path
- A global reverse proxy routes HTTP traffic by path. Each artifact declares its paths in `.replit-artifact/`
- For the Expo app: use `$REPLIT_EXPO_DEV_DOMAIN` directly
- artifact.toml. Routes are matched most-specific-first.

7. Getting the Source & Running Locally

Step 1 — Download the source

In Replit, click the three-dot menu ("⋮") in the file panel and choose "Download as ZIP". Extract the zip on your machine.

```
git clone https://github.com/<your-org>/rewlo.git && cd rewlo
```

Step 2 — Install prerequisites

- Node.js 20 or 24 (<https://nodejs.org>)
- pnpm 9+ !' npm install -g pnpm
- Expo Go app on your iOS/Android device (for quick testing)
- Optional: Xcode 15+ (iOS simulator on Mac) or Android Studio

Step 3 — Install dependencies

Step 4 — Environment variables

```
DATABASE_URL=postgres://dev:password@localhost:5432/rewlo
```

```
SESSION_SECRET=any-random-string-here
```

Create a `.env` file at the project root with at minimum:

```
pnpm --filter @workspace/mobile run dev
```

The mobile app itself does NOT need `DATABASE_URL` — only the API server does. You can skip the database entirely if you only want to run the Expo app.

```
# In a second terminal (optional — API server):
```

```
pnpm --filter @workspace/api run dev
```

Step 5 — Run the mobile app

Expo will print a QR code. Scan it with the Expo Go app to open RewLo on your device. Press "i" for iOS simulator or "a" for Android emulator.

Step 6 — Sign in (Early Access Preview)

```
pnpm --filter @workspace/mobile run typecheck
```

Any non-empty email and password will log you in — WalletContext accepts any credentials during the preview phase. No account creation is needed.

```
# Full workspace check:
```

Typecheck (no build needed)

```
pnpm run typecheck
```

Regenerate API client (if you change the OpenAPI spec)

```
pnpm run generate-api-client
```

Push DB schema changes

```
pnpm --filter @workspace/api run push # dev only
```

8. Building for App Store (iOS)

```
pnpm install --save-dev eas-cli
eas login
```

```
eas build --platform ios # cloud build
```

```
eas submit --platform ios # submit to App Store Connect
```

The project uses EAS (Expo Application Services) for App Store builds. You need an Expo account and Apple Developer account.

- `eas.json` at `artifacts/mobile/eas.json` controls build profiles
- Bundle ID: `com.rewlo.app`
- Requires Apple Team ID and App Store Connect API key configured in EAS
- `ITSAAppUsesNonExemptEncryption`: `false` already set (no export compliance needed)

9. Support & Contacts

Support email: support@rewlo.io

App bundle ID: `com.rewlo.app`

Expo slug: `rewlo`

For questions about the codebase, reach out via the Replit project or the team Slack.